Supplementary material

This supplement provides additional definitions, denominators, diagnostics, and reporting information for the nationwide study of therapeutic ERCP in Brazil's Unified Health System, 2021–2025. It reports prespecified analyses without converting unavailable results into null findings.

Policy, coding, and observation chronology

Four national landmarks framed the observation window: the final CONITEC recommendation on July 4, 2019; Portaria SCTIE/MS no. 39 on July 24, 2019; Portaria GM/MS no. 3,728 on December 22, 2020; and the presence of therapeutic ERCP code 0407030255 in the January 2021 SIGTAP archive. The code was absent from the December 2020 archive. January 2021 therefore marks the beginning of a homogeneous coded observation window, not the beginning of ERCP practice in Brazil. Hospitals active in that month were treated as left-censored prevalent users.

Data sources and acquisition boundaries

Supplementary Table 1. Data sources and acquisition boundaries.

Source	Custodian	Analytical role	Sharing boundary
SIH/SUS hospitalisation and procedure-detail records	DATASUS, Ministry of Health	AIH unit, procedure code, diagnoses, residence, provider, in-hospital disposition	Restricted individual rows are not redistributed
CNES-PF	DATASUS	Time-aligned hospital eligibility and capacity context	Public-source acquisition instructions only
Population and municipality geometry	IBGE	Adult denominators, municipal identifiers, geometry, audited city-seat anchors	Publicly documented acquisition route
IVS 2010	Ipea-linked official source	Static municipal ecological context	No individual socioeconomic interpretation
Supplementary-insurance coverage	ANS	Municipal contextual measure	Contextual, not individual insurance status
OpenStreetMap extract	OpenStreetMap contributors	Road graph (brazil-260822.osm.pbf)	Licence and acquisition route documented
GraphHopper	Local reproducible service	ercp_car route and contour profile	Version/configuration evidence must be mirrored before final integrity
CONITEC, Portarias, SIGTAP	Ministry of Health	Policy and code chronology	Official sources and access dates documented

AIH, Autorização de Internação Hospitalar; ERCP, endoscopic retrograde cholangiopancreatography.

The study used SIH/SUS hospitalisation and procedure-detail records, CNES hospital data, IBGE population and municipality geography, the 2010 municipal IVS, supplementary-insurance coverage, OpenStreetMap road data, GraphHopper routing, and official policy sources. Record-level SUS files, credentials, and unredacted small cells are not distributed.

Code lists, linkage, and cohort construction

Procedure-detail rows containing 0407030255 were collapsed by competence month, provider CNES identifier, and AIH number before linkage. Cohort A contained 91,175 unique AIHs, all exactly linked; 90,146 involved adults. Among 33,888 AIHs with K80.3, K80.4, or K80.5 in a valid diagnostic field, 33,506 involved adults. Nine adult records were excluded because the principal diagnosis belonged to the C23 or C24 family, leaving 33,497 Cohort B AIHs. Duplicate counts were zero, and Cohort B was a strict subset of Cohort A.

Cohort B is an administrative diagnostic phenotype. No chart adjudication, sensitivity or specificity study, or externally validated performance estimate was available.

Variable availability and missing-data handling

Identifiers, linkage keys, procedure presence, first observed coded use, and primary outcomes were not imputed. Municipalities without a valid road anchor remained unknown rather than uncovered. Race or colour was represented by an explicit missing category in the prespecified outcome design because recording may be institutionally patterned. Variables lacking adequate semantic or analytical support were excluded rather than imputed after analysis.

The Aim 4 complete-context analysis included 30,900 of 33,497 Cohort B AIHs, 378 in-hospital deaths, and 360 hospitals. A variable-level decomposition of the 2,597 excluded records was unavailable. State-, year-, hospital-, and cohort-specific missingness patterns were therefore not inferred from aggregate totals.

Cross-aim estimands, denominators, and analysis boundaries

Supplementary Table 2. Cross-aim estimands, denominators, and analysis boundaries.

Component	Cohort / unit	Denominator	Estimand or descriptive target	Inference level	Missingness / unknown rule	Reporting status
Aim 1 observed uptake	Cohort A hospitals	445 hospitals ever observed	First month of observed coded therapeutic ERCP use	Descriptive	January 2021 active hospitals are left-censored prevalent users	Reported
Aim 1 maintenance	Cohort A hospitals	390 with a completed first 12-month window	Activity in at least 6 months of the first completed 12-month window	Descriptive	Incomplete windows excluded from the risk set	252/390 (64.6%); reported
Aim 1 first-observed-use model	Eligible hospital-months	Frozen CNES risk set	Associational model of first observed coded use	Associational	No causal adoption interpretation	Associational; limited events per parameter
Aim 2 ecological equity	Cohort B municipality-years	27,825 municipality-years	Adult-standardised treated-use gradient across ranked IVS	Ecological associational	Static 2010 IVS; no individual interpretation	SII and RII reported; modified-Poisson inference DOWNGRADE
Aim 2 realised routes	Cohort B routed AIHs	33,321 routed AIHs	Flow-weighted route-time distribution among treated flows	Descriptive	Unrouted flows excluded from route summaries; no unmet-need inference	Reported
Aim 2 potential reach	Adult municipality-year population	801,100,903 adult person-years in anchor audit	Population within 120 and cumulative 180 road-minutes of observed provider municipalities	Descriptive modelled opportunity	Invalid anchors remain unknown; one municipality unknown in 2025	National component reported; regional and vulnerability-gap components NOT_EVALUATED
Aim 3 patient-flow network	Observed residence-to-provider flows and pooled municipal projection	3,338 nodes; 166,959 shared-destination edges	Destination and shared-destination structure	Descriptive	Public cells below 5 suppressed; no patient-to-patient interpretation	Reported
Aim 3 service areas	Residence municipality-year records	5,565 municipal IVS records for contextual summaries	Modal observed destination and destination share	Descriptive	Suppressed flows remain suppressed; no need or appropriateness inference	Reported
Aim 3 structural resilience	Potential-access matrix / adults	8,723,036 matrix rows; 142,299 reachable rows	Adults losing an alternative provider under targeted hub removal	Descriptive structural stress test	Complete-case and upper-bound handling retain unknown anchors	Reported as a structural stress test; not a real-world counterfactual
Aim 4 supportive outcome	Cohort B complete-context AIHs	30,900 AIHs; 378 deaths; 360 hospitals	Standardised risk contrast at hospital-volume p90 versus p10	Associational/supportive	Complete-context design; no imputation-based alternative was used	Bootstrap DOWNGRADE; formal CI NOT_EVALUATED
Planned staggered-uptake analysis	Hospital-month design	No pre-first-observed-use hospital-months	Planned quasi-causal family	Not a release result	Structural absence of pre-period observations	Not reported because pre-period observations were unavailable

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Observed uptake and maintenance diagnostics

Observed uptake was defined as first observed coded use rather than true adoption. Maintenance required coded activity in at least six months of the first complete 12-month window; only hospitals with a fully observable window entered the denominator. The first-observed-use model had limited events per parameter and was interpreted as associational.

Ecological equity, routing, and potential reach

The 2010 municipal IVS was carried forward as a static ranked ecological exposure through 2025. The RII was 0.55 and the SII was -2.56 treated AIHs per 100,000 adults. These estimates do not identify individual socioeconomic effects.

Realised route estimates were available for 33,321 Cohort B AIHs. The flow-weighted median was 23.2 minutes, the 90th percentile was 213.4 minutes, and 20.7% exceeded 120 minutes. Municipal anchors approximate locations rather than patient addresses or observed journeys.

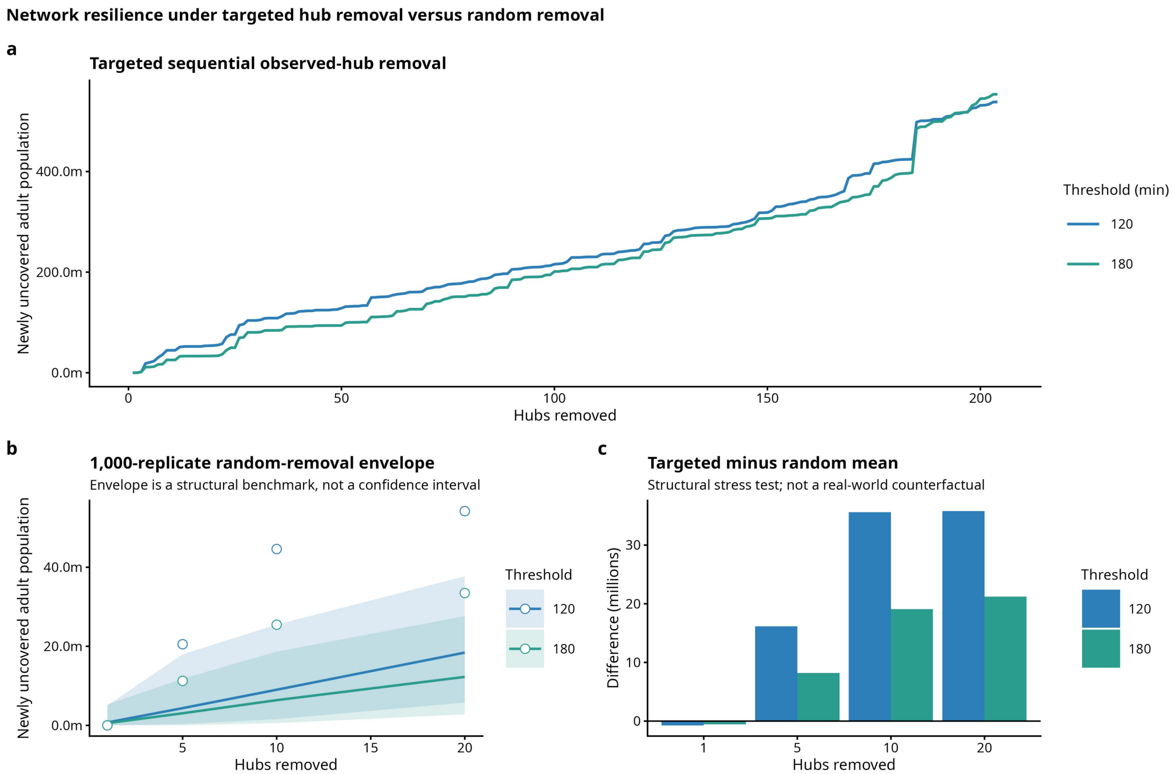
All 456 provider-threshold routing entries were completed. IBGE city-seat anchors were substituted for municipalities 314800, 330010, and 510790 only after their original anchors returned Point not found. Twenty-nine timeout-only municipalities retained their original anchors with a longer runtime. Because raw contours were non-nested for 58 providers, cumulative 180-minute reach used the union of the 120- and 180-minute polygons. Adult person-year anchor coverage was 99.999504%; one municipality remained without a valid anchor in 2025. National 120-minute potential reach increased from 71.9% to 82.1%, and cumulative 180-minute reach from 81.2% to 90.4%. Regional coverage and a municipal vulnerability-gap analysis were not evaluated because the required region mapping and municipality-level IVS join were unavailable.

The prespecified modified-Poisson model produced some standardised potential-coverage predictions above 1. This model family was therefore downgraded and its predictions were excluded from formal inference. The descriptive national coverage series remains reported separately.

Patient-flow network, service areas, privacy, and structural resilience

The pooled municipal projection contained 3,338 nodes and 166,959 shared-destination edges. Public network displays suppressed counts below five, and suppressed values were not imputed. Centrality and service-area summaries do not establish referral quality, capacity, or appropriateness.

Structural resilience was examined by sequential removal of prespecified high-in-strength hubs and comparison with seeded random removal at k=1, 5, 10, and 20. Each threshold-scenario used 1,000 replicates with seed 20260830. Random 2.5th–97.5th percentile envelopes are simulation benchmarks, not confidence intervals. At 180 minutes, targeted removal of 5, 10, and 20 hubs newly uncovered 11.3, 25.5, and 33.5 million adults, compared with random means of 3.1, 6.4, and 12.3 million. The stress test does not model facility closure, capacity redistribution, behavioural adaptation, or actual rerouting.



Supplementary Figure 1. Structural hub-removal stress test. Targeted high-in-strength hub removal is compared with seeded random-removal benchmarks. This is a structural stress test, not a prediction of real-world closure or rerouting.

Supportive in-hospital outcome analysis

The outcome was in-hospital death. Exposure was trailing 12-month all-indication unique Cohort A ERCP AIH volume at the treating hospital, represented by a four-knot restricted cubic spline. The complete model included prespecified patient, hospital, contextual, provider-state, and calendar-month terms. ICU use, length of stay, reimbursement, and other post-exposure variables were excluded. Reimbursement is a payment field, not a measure of cost.

Separation diagnostics triggered two prespecified bias-reduced estimators. Standardised risks at volume p10 and p90 were 0.014341850368641714 and 0.014306146454349527; the risk difference was -0.000035703914292186875 and the risk ratio was 0.9975105085205566. These are associational point estimates. Their numerical proximity does not establish no association or equivalence.

The hospital-cluster bootstrap used 2,000 replicates and seed 20260830. Formal inferential reporting required at least 1,900 valid replicates; 1,800–1,899 would support exploratory reporting, and fewer than 1,800 required downgrade. Only 618 AS_mean and 617 MPL_Jeffreys replicates were valid. Formal percentile intervals were therefore not evaluated, and bootstrap quantiles are not presented as confidence intervals. Among failed replicates, AS_mean had 726 nonconvergence and 656 rank-deficient failures; MPL_Jeffreys had 726 nonconvergence, 656 rank-deficient failures, and one unclassified failure. These categories concern computational validity rather than clinical heterogeneity. No reduced model, altered seed, lower replicate target, or substitute interval was used.

Planned quasi-causal analysis

No hospital-months preceded first observed coded use, so the prespecified causal eligibility condition could not be met. Quasi-causal estimates, negative controls, and associated robustness analyses are not presented. This absence is not a null finding.

Additional figure and tables

Supplementary Table 3. Ecological equity, travel, potential reach, and network measures.

Measure	Value	Denominator	Evidence	Key limitation
IVS relative index of inequality	0.55	27,825 municipality-years	associational	IVS is an ecological contextual exposure and does not identify individual socioeconomic effects.
IVS slope index of inequality	-2.56 treated AIHs per 100,000 adults	27,825 municipality-years	associational	IVS is an ecological contextual exposure and the association is not causal.
Flow-weighted median realised route time	23.2 minutes	33,321 AIHs with a route	descriptive	Municipal anchors approximate residence and provider locations; this is realised treated flow, not direct patient travel.
Flow-weighted 90th percentile realised route time	213.4 minutes	33,321 AIHs with a route	descriptive	Municipal anchors approximate residence and provider locations; this is realised treated flow, not direct patient travel.
Realised treated flow above 120 minutes	20.7%	33,321 AIHs with a route	descriptive	This describes routed treated flows and cannot be interpreted as population access or unmet need.
Adult population within 120 road-minutes in 2021	71.9%	157,421,555 adults	descriptive	Modelled potential road access does not measure realised access, capacity, quality, or need.
Adult population within 180 road-minutes in 2021	81.2%	157,421,555 adults	descriptive	The cumulative 180-minute estimand unions raw 120- and 180-minute contours when GraphHopper approximation is non-nested.
Adult population within 120 road-minutes in 2025	82.1%	163,078,720 anchored adults	descriptive	One 2025 municipality lacks a road anchor; anchored-denominator and bounds must be reported.
Adult population within 180 road-minutes in 2025	90.4%	163,078,720 anchored adults	descriptive	One 2025 municipality lacks a road anchor; the cumulative geometry rule and bounds must be retained.
Adult population road-anchor coverage	99.9995%	801,100,903 adult person-years	descriptive	Municipalities without anchors remain unknown and are never recoded as uncovered.
Aim 2 primary-family inferential status	DOWNGRADE	—	associational	Modified-Poisson standardized potential-coverage predictions exceeded probability 1; the model was not rerun or relaxed.
Pooled patient-flow projection nodes	3,338 municipalities	—	descriptive	The patient-flow network reflects observed treated flows, not population need or referral appropriateness.
Pooled patient-flow projection edges	166,959 edges	—	descriptive	Projection edges encode shared performing destinations and are not direct patient-to-patient links.
Potential-access matrix rows	8,723,036 rows	—	descriptive	This is a modelled potential-access matrix and contains no capacity or realised-use inference.
Reachable potential-access matrix rows	142,299 rows	8,723,036 potential-access matrix rows	descriptive	Reachability is potential road access, not observed service use or capacity.
IVS municipal-total records selected	5,565 municipalities	—	descriptive	The 2010 IVS is a static ecological contextual exposure carried across 2021-2025.

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Supplementary Table 4. Supportive outcome-analysis measures and inferential limitations.

Measure	Value	Denominator	Evidence	Key limitation
Aim 4 complete primary analysis rows	30,900 AIHs	33,497 Cohort B unique AIHs	associational	Complete-context restriction and residual confounding limit generalisability and causal interpretation.
Aim 4 in-hospital death events	378 deaths	30,900 Aim 4 analysis AIHs	associational	Administrative in-hospital death does not capture post-discharge mortality.
Aim 4 hospitals	360 hospitals	30,900 Aim 4 analysis AIHs	associational	Hospital-level associations remain susceptible to residual confounding and referral selection.
Aim 4 bootstrap validity status	DOWNGRADE	2,000 expected replicates	associational	The prespecified bootstrap validity criterion was not met; no reduced model or substitute interval was used.
Valid AS_mean bootstrap replicates	618 replicates	2,000 expected replicates	associational	Valid-replicate yield was below the prespecified threshold; inferential intervals are downgraded.
Valid MPL_Jeffreys bootstrap replicates	617 replicates	2,000 expected replicates	associational	Valid-replicate yield was below the prespecified threshold; inferential intervals are downgraded.
Planned quasi-causal analysis	Not reported	—	associational	No pre-first-observed-use hospital-months were available; quasi-causal estimates and negative controls are not reported.

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RECORD reporting checklist

Supplementary Table 5. RECORD reporting checklist.

RECORD item group	Locator	Status
Routinely collected data source and setting	Methods: Data sources; the supplementary data-sources section	Reported
Code/algorithm-based population selection	Methods: Cohorts; the supplementary cohort-construction section	Reported
Validation of codes/algorithms	Discussion limitations; the supplementary cohort-construction section	No chart validation available; explicitly stated
Linkage methods and quality	Methods: Cohorts; the supplementary cohort-construction section	Exact linkage rule reported; reason-specific intermediate breakdown unavailable
Data cleaning and deduplication	Methods: Cohorts; the supplementary cohort-construction section	Reported
Data access and privacy	Data availability; the supplementary sections 'Data sources and acquisition boundary' and 'Aim 3 patient-flow network, service areas, privacy, and resilience'	Reported
Flow of records and exclusions	Figure 1; the supplementary sections 'Code lists, linkage, and cohort construction' and 'Variable availability and missing-data handling'	Aggregate flow reported; reason-specific intermediate breakdown unavailable
Extent of database access for replication	Code/Data availability; the supplementary reproducibility and declarations section	Reported with restricted-row boundary

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STROBE reporting checklist

Supplementary Table 6. STROBE reporting checklist.

STROBE item	Locator	Status
Title/abstract design	Title; Structured abstract	Reported
Background and objectives	Introduction	Reported
Study design, setting, dates	Methods: Study design	Reported
Participants and eligibility	Methods: Cohorts; the supplementary cohort-construction section	Reported
Variables, outcomes, exposures, confounders	Methods by aim; the supplementary estimand map and Aim 4 section	Reported
Data sources and measurement	Methods: Data sources; the supplementary sections on data sources and the aim-specific methods	Reported
Bias	Methods: Missing data; Discussion limitations	Reported
Study size	National administrative census; the supplementary estimand map	Reported; no sampling power calculation used
Quantitative variables and statistical methods	Methods by aim; the supplementary estimand map and aim-specific sections	Reported
Missing data	Methods; the supplementary missing-data section	Reported within frozen local evidence limits
Participants and descriptive data	Results; Figures 1–2; Tables 1–2	Reported
Outcome and main results	Results; Figures 3–6, Supplementary Figure 1, and Supplementary Tables 5–6	Reported with downgrades
Other analyses	the supplementary aim-specific and quasi-causal eligibility sections	Reported; planned quasi-causal estimates excluded
Key results, limitations, interpretation, generalisability	Discussion	Reported
Funding	Declarations	Pending author metadata

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